Thermodynamics and Statistical Physics

8th Exercise Class

Canonical ensemble; Kinetic theory of gases; Equipartition theorem

Exercises for the Class

1. Assume that a system can be divided into N non-interacting subsystems. Show that in this case the partition function can be calculated as

$$Z = \prod_{k=1}^{N} z_k,$$

where z_k is the partition function of the kth subsystem. What is the result if the N subsystem corresponds to identical, indistinguishable particles? (Indistinguishable means, that by permuting the particles, we get the same microstate.)

2. The Helmholtz free energy.

   (a) Show that the mean energy of a system with partition function Z is

$$U = \langle E \rangle = -\frac{1}{Z}\frac{\partial Z}{\partial \beta},$$

   where $\beta = 1/(k_{\mathrm{B}}T)$.

   (b) Show that the Helmholtz free energy can be expressed by the partition function as

$$F = -k_{\mathrm{B}}T \ln Z.$$

3. A system consists of $N \gg 1$ non-interacting, distinguishable two-level atoms. Each atom can be in one of two states 0 and 1 with energies $E_0 = 0$ and $E_1 = \varepsilon$. The system is in equilibrium with the heat bath at temperature T.

   (a) Find the partition function and the Helmholtz free energy.

   (b) Calculate the internal energy, the entropy, and the heat capacity as functions of temperature.

4. In quantum mechanics a particle of mass m confined to a one-dimensional infinite potential well of width L has quantized energy states

$$E_n = \frac{n^2 h^2}{8mL^2},$$

where the quantum number n is restricted to positive integers and h is the Planck constant.

   (a) Write the partition function for a single particle in one dimension in the form of a sum over all possible quantum numbers n.

   (b) Change the sum to an integral and thereby show that for this system

$$Z = L\sqrt{\frac{2\pi m k_{\mathrm{B}} T}{h^2}}$$

   (c) Use the result of the Exercise 1, and calculate the partition function of N non-interacting, indistinguishable particles in a three-dimensional box.

5. In one second, $5 \cdot 10^{23}$ nitrogen molecules collide with a flat wall of 8 cm^2 area. The average value of the velocity component of the molecules perpendicular to the wall is 300 m/s.

   (a) What is the pressure on the wall?

   (b) Calculate the temperature of the gas and the average kinetic energy per molecule (translational and rotational). Use the approximation: $\langle |v_x| \rangle \approx \sqrt{\langle v_x^2 \rangle}$.

   (c) Estimate the average distance between the molecules.

6. The air temperature in a weekend house is increased from 10 °C to 25 °C by heating. Due to the ventilation gaps at the windows of the house, the internal air pressure is the same as the outside pressure all the time. Does it increase, and if so, by how much:

   (a) the average kinetic energy of the air particles,

   (b) the total internal energy of the air in the house?

7. What could be the gas whose average thermal velocity increases from 400 m/s to 500 m/s when the temperature is raised by 100 K?

Homework to Be Submitted

1. A paramgnetic system contains N non interacting, distinguishable magnetic atoms within a volume V. Each atom can have magnetic moment of $\pm\mu$. Placing this material in an external magnetic field B, the magnetic moments can be parallel or antiparallel to the direction of the magnetic field.

   (a) Calculate the average magnetic moment per atom.

(b) Find the magnetization of the system.

(c) Determine the partition function of the system.

(d) Give the Helmholtz free energy.

(e) Calculate the magnetization from the free energy and compare it to the result find in part (b).

(f) How does the magnetization depend on the temperature if the magnetic field is weak and temperature is high?

Hint: The magnetic energy of a magnetic moment $\boldsymbol{\mu}$ in a magnetic field $\boldsymbol{B}$ is $E = = -\boldsymbol{\mu}\boldsymbol{B}$.

In the present system the Gibbs energy is equal to the Helmholtz free energy.

Homework for Practice

1. The possible particle energies of a system of particles are $0, \varepsilon, 2\varepsilon, \ldots, n\varepsilon, \ldots$.

   (a) Give the partition function.

   (b) Compute the average energy of the particles.

   (c) Calculate the average occupation number.

   (d) Find the limiting value of the average energy and the average occupation number when ε is much smaller than $k_{\mathrm{B}}T$.

2. A thermally insulated container is divided into two sections by a thermally insulating wall. One section contains 0.1 mol of oxygen gas, and the other contains 0.2 mol of neon gas. The average thermal speeds of the translational motion of the oxygen molecules and the neon atoms are equal. We remove the partition. How many times the initial absolute temperature of the oxygen is the resulting common temperature?

3. There is an ideal gas consisting of N molecules of mass m in a spherical container of radius R. Assuming that the molecules move randomly in all directions but with equal speeds, derive the fundamental equation of the kinetic theory of gases, that is, determine the pressure of the gas.